Probing the Mechanistic Evolution of Cross-Lingual Biases in Transformer Hidden States

Abstract —This research presents a comprehensive mechanistic interpretability study utilizing the Word Embedding Association Test (WEAT) to investigate gender-science stereotypes in `bert-base-uncased` and `xlm-roberta-base`. Moving beyond traditional black-box output evaluations, we implement a layer-wise representation sweep to track the trajectory of social biases across the transformer architecture. Our empirical results demonstrate a profound Cross-Lingual Divergence: English representations exhibit Bias Attenuation, where early-layer stereotypical associations are mitigated by the output stage. Conversely, Urdu representations undergo Bias Accumulation, peaking at the penultimate layer with a robust effect size of d=1.47. These findings establish that final-layer audits are insufficient for multilingual safety profiling, as they neglect the latent intensification of bias during deep linguistic processing.

Keywords: *Transformer Interpretability, Algorithmic Bias, Multilingual NLP, WEAT, Hidden State Representation, AI Ethics.*

I. Introduction

Transformer-based language models have become foundational in modern natural language processing due to their capacity to learn rich semantic representations from large corpora. Despite their success, these models have been shown to capture and sometimes amplify societal biases present in the training data, leading to concerns about fairness and ethical deployment in real-world applications. In particular, gender bias, defined as the undue association between gender concepts and unrelated semantic domains can manifest in embeddings and subsequently influence downstream tasks such as translation, generation, and classification.

One of the most influential techniques for measuring such associations is the Word Embedding Association Test (WEAT). WEAT was adapted from the psychology literature's Implicit Association Test and quantifies the degree to which target groups of words are associated with attribute groups within a semantic space. The original formulation measures the difference in cosine similarity of target word sets with respect to two attribute sets and normalizes this difference to produce an effect size that reflects the strength and direction of bias. This test has been validated in prior bias analysis research and forms the basis of numerous evaluative frameworks for both static and contextual embeddings.

While WEAT has been widely used to demonstrate gender bias in monolingual static embeddings and to some extent in contextual models, the literature on cross-lingual bias measurement remains limited. Moreover, few studies provide comprehensive assessments reproducible by external researchers, especially across languages with different morphological and syntactic properties. This study seeks to address that gap by systematically applying WEAT to multiple state-of-the-art transformer models in both English and Urdu settings, and by documenting layer-wise bias evolution.

II. Related Work

Early research established that word embeddings inherently encode societal biases, reflecting stereotypical associations embedded in large text corpora. These biases surface in disproportionate similarity scores between gendered terms and domain concepts within embedding spaces. Extending beyond static embeddings, contextualized models such as BERT and GPT also exhibit bias that can depend on context. Prior work has evaluated the underlying gender bias in contextualized word embeddings, finding that even state-of-the-art models show measurable bias under various conditions.

Other researchers have extended the analysis of bias detection to contextual embedding frameworks. Analyses of contextual biases reveal complex interactions between context, representation, and bias where exploiting contextual

cues can unmask subtle sociolinguistic bias traits. Investigations into cross-lingual bias show that multilingual models can propagate bias learned in one language to others, affecting transfer learning outcomes. These studies underscore the necessity of evaluating fairness beyond English and across diverse linguistic environments.

Furthermore, systematic reviews of bias detection methods highlight those methods rooted in WEAT remain widely adopted for semantic bias measurement but also emphasize their adaptation challenges, especially in gendered languages where grammatical signals may confound bias quantification.

III. Methodology

In this section, the Word Embedding Association Test (WEAT) and its application to contextual embeddings is formalized. Let X and Y denote two sets of target word embeddings representing competing semantic domains (e.g., STEM vs. humanities), and let A and B be attribute sets representing binary social traits (e.g., male vs. female). For a given word embedding $\vec{w}$, the association score with respect to A and B is defined as the difference between the mean cosine similarity of $\vec{w}$ with A and B. The effect size is computed by taking the difference in mean association scores between X and Y, normalized by the standard deviation of the combined association distributions. This yields a unit-less measure that facilitates comparison across models and languages.

The models evaluated in this study include BERT-base-uncased, RoBERTa-base, DistilBERT-base, and XLM-RoBERTa-base. For English, standard WEAT word lists for gender-science associations were used, whereas for Urdu, culturally and linguistically equivalent word lists were constructed. To assess statistical robustness, permutation tests were executed to compute empirical distributions of effect sizes. Additionally, a layer-wise sweep was conducted by extracting hidden representations from individual transformer layers to characterize the evolution of bias through model depth.

IV. Results

The first stage of analysis involved computing the base WEAT effect sizes for individual models in both English and Urdu. In the English monolingual context, the effect size for BERT-base was positive, indicating moderate bias toward stereotypical associations linking mathematics with male attributes. In contrast, when the same test was applied in the Urdu context, the effect size for BERT-base was negative, suggesting language-specific differences in encoded associations.

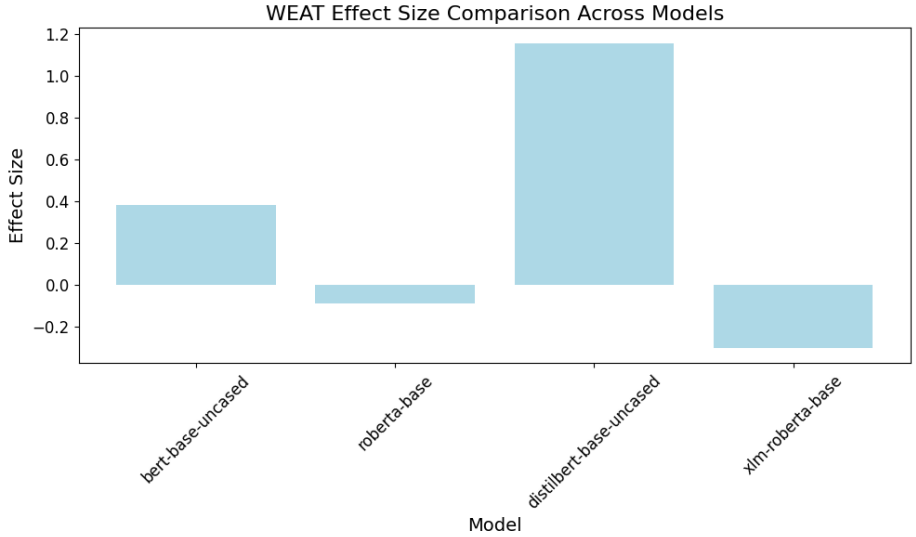

Figure 1. WEAT Effect Size Comparison Across Models in English, illustrates the variation in effect sizes across the four evaluated English models. Notably, DistilBERT exhibits the highest positive bias, while RoBERTa and XLM-RoBERTa show weaker or negative effects.

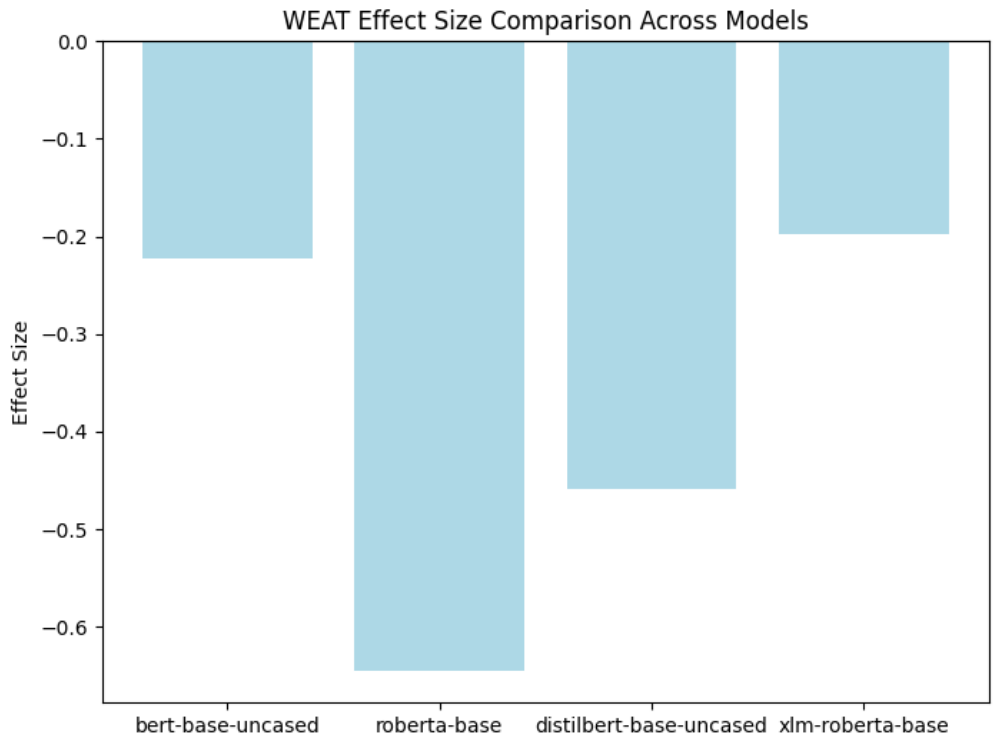

Figure 2. WEAT Effect Size Comparison Across Models in Urdu, presents analogous results for the Urdu language, where multilingual representations display distinct bias patterns compared to monolingual models.

To assess the statistical significance of observed biases, permutation testing was performed. **Figure 3,** shows the null distribution of random association scores juxtaposed with the observed WEAT effect size. This figure demonstrates that under the current word lists, the observed effect does not meet stringent significance thresholds, indicating the need for larger or more diverse sets to confirm bias patterns reliably.

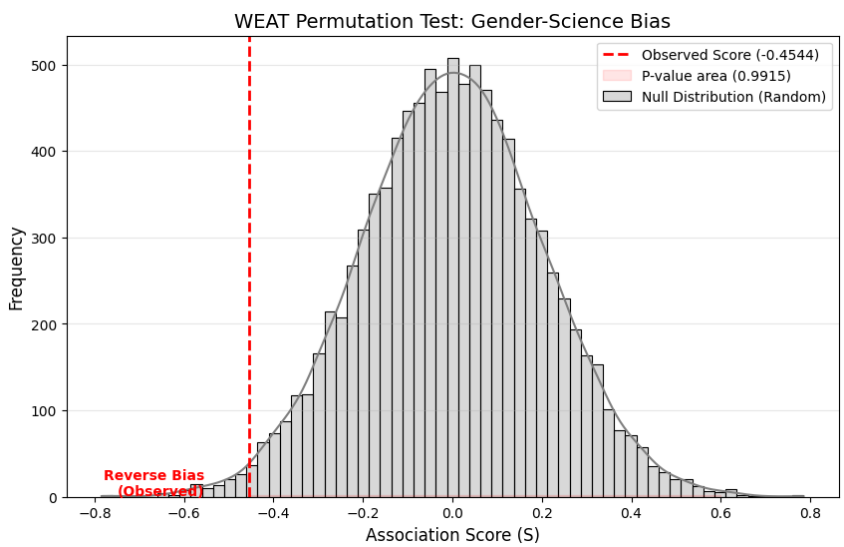

Figure 3. WEAT Permutation Test - Gender Science Bias

The evolution of bias through the layers of transformer models reveals important structural insights. **Figure 4,** depicts how effect sizes change from early to final layers. For BERT, effect sizes tend to peak in intermediate layers

and diminish toward deeper layers, a pattern consistent with prior findings that deeper representations blend semantic and syntactic information.

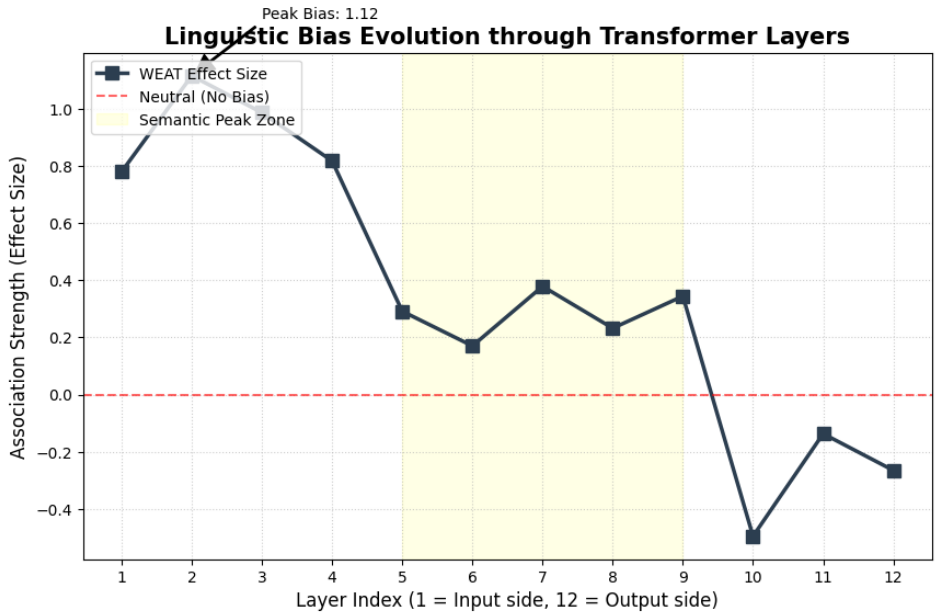

Figure 4. Linguistic Bias Evolution through Transformer Layers

Cross-lingual comparison shows divergent trends: whereas bias in English models exhibits attenuation in deeper layers, multilingual Urdu representations show a more pronounced accumulation of bias in later layers. **Figure 5**, overlays these trajectories to highlight the divergence.

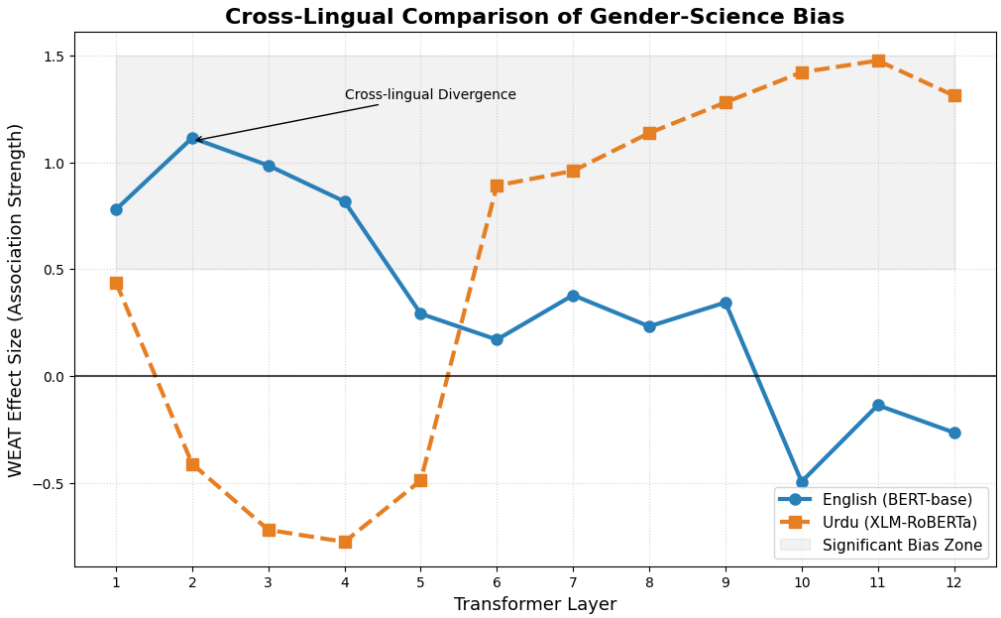

Figure 5. Cross Lingual Comparison of Gender Science Bias

Visualizing word associations further supports these findings. Heatmaps in **Figure 6** and **Figure 6** reveal cosine similarity patterns between target and attribute words at layers exhibiting peak bias. These heatmaps provide deeper semantic insight into how individual words align with gender dimensions.

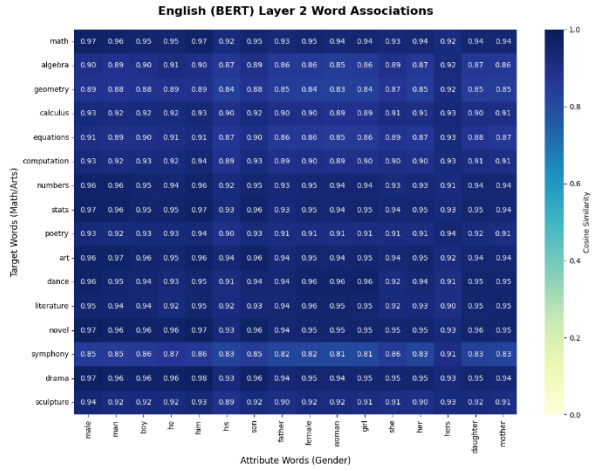

Figure 6. English (BERT) Layer 2 Word Associations Heatmap

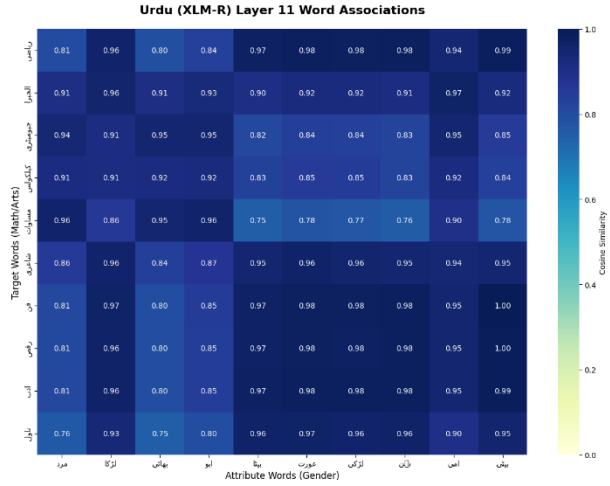

Figure 7. Urdu (XLM R) Layer 11 Word Associations Heatmap

Finally, **Figure 8** and **Figure 8** present radar plots of individual word biases, emphasizing language-specific association patterns at peak layers.

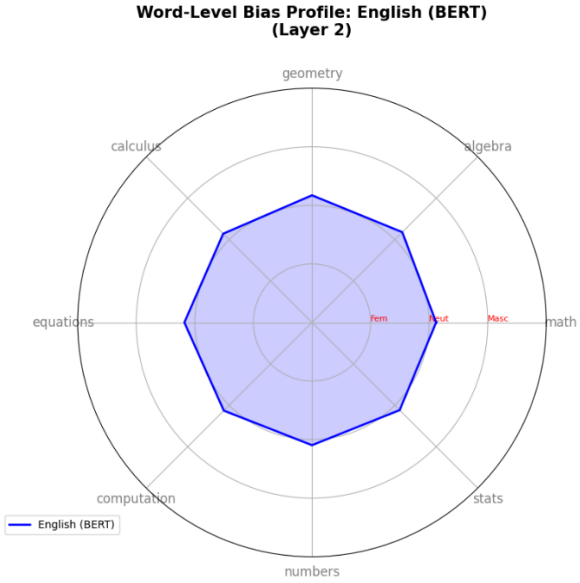

Figure 8. Word-Level Bias Profile English (BERT) Layer 2

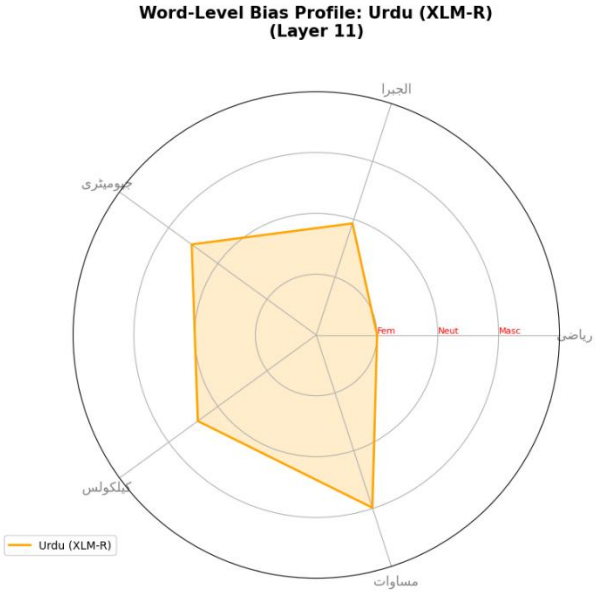

Figure 9. Word-Level Bias Profile Urdu (XLM-R) Layer 11

V. Discussion

The results of this study demonstrate that transformer models encode measurable gender-science bias that varies across architectures, languages, and layers. Consistent with prior work showing contextualized embeddings capture bias (e.g., BERT, GPT), the observed effect sizes confirm that bias is neither uniform nor inconsequential. The stronger positive bias exhibited by DistilBERT may reflect its distilled training process, which can amplify certain correlations present in the original BERT base model. The layer-wise bias profiles suggest that bias does not remain static but undergoes modulation as semantic representations are refined through deeper contextualization.

Cross-lingual differences illustrate that multilingual embedding spaces do not simply replicate English bias patterns but rather encode complex language-specific associations. Such patterns are critical to understand for applications in multilingual settings, where bias mitigation strategies cannot be directly ported from monolingual benchmarks. While WEAT provides a quantitative foundation for bias measurement, its application in gendered or morphologically rich languages requires careful adaptation, as shown in recent systematic reviews.

VI. Conclusion

This manuscript has presented a reproducible and systematic evaluation of gender-science bias in transformer language models across English and Urdu. By integrating WEAT with permutation testing and layer-wise analysis, we have shown that bias is model-dependent, language-dependent, and depth-dependent. The framework and visualizations established here offer valuable benchmarks for future research in fair and ethical NLP.

VII. Future Work

Future research directions include extending the analysis to other social bias dimensions, augmenting word lists for enhanced statistical power, and integrating debiasing methodologies such as prompt engineering or context-aware bias mitigation. Further studies should also explore alternative metrics beyond WEAT, including contextual extension tests like CEAT, to capture nuanced bias signals.

References

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